

1) Construye un cuadruplo RC en modo que cumpla con:

$$-Y_{21} = \frac{\$ (9+1)}{\$+2} \quad \text{y} \quad Y_{22} = \frac{\$ (\$+2+25)}{\$+2} \quad \text{tal que} \quad \frac{V_2}{V_1} = \frac{\$+1}{(\$+2+25)} \quad \text{con } I_2=0$$

10) Observa estructuras de hielo y montes coeantes de perimetros

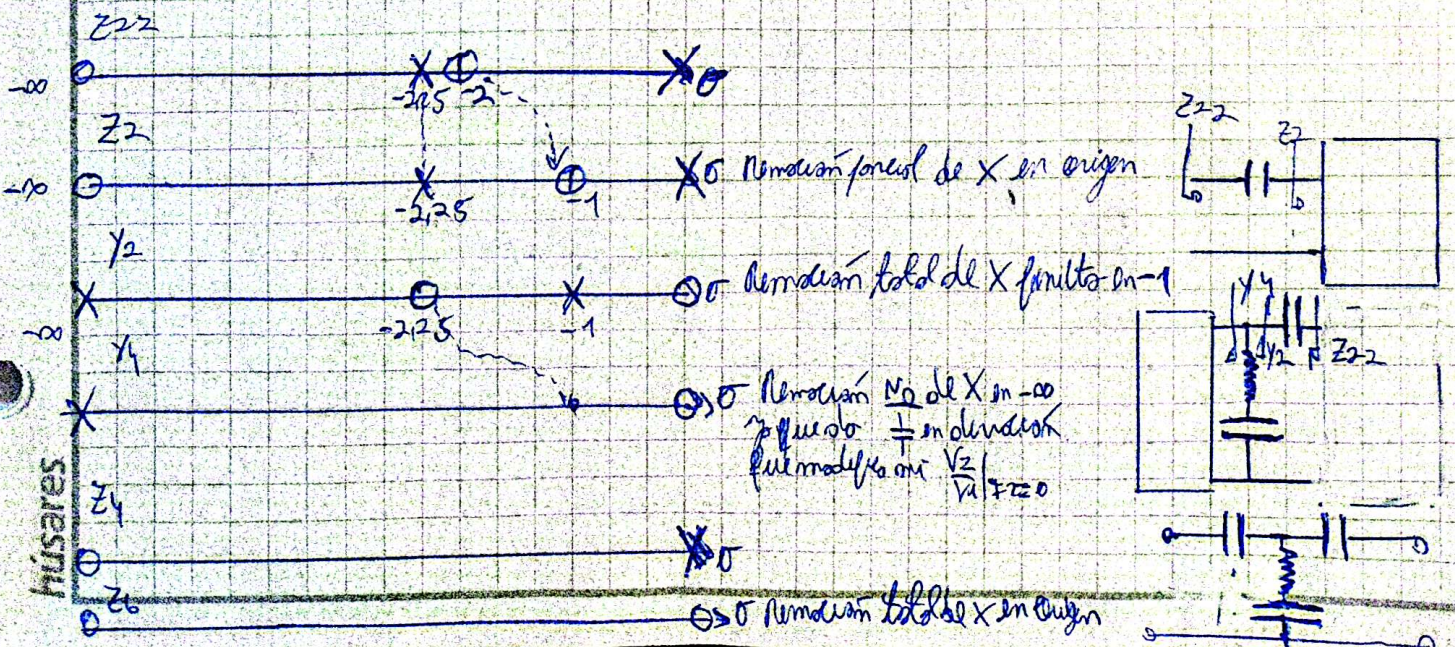
→ De $-\frac{1}{2}x = \frac{g(g+1)}{g+2}$, take "Euros" en $g=0 \wedge g=-1$ y pda en $g=-2$

→ Die $f_{22} = \frac{f(s+2, 2.5)}{s+2}$, keine "Zeros" an $s = 0.1$ & $z = -2.25$ und an $s = -2$

Dado que la luz solar y "zonas" sobre eye o yellow $\Rightarrow$ sintaxis descriptiva

$$\rightarrow -\frac{y_1}{y_2} = \frac{\frac{\$ (8+1)}{\$ 2}}{\frac{\$ (8+2.25)}{\$ 2}} = \frac{\$ 1}{\$ 2.25} = \frac{V_2}{V_1} \Big|_{I_2=0}$$

29) Aplica métodos gráficos, debes comenzar con $V_{22} = \frac{1}{Z_2}$ puesto que los
ceros de V_{22} producen polos en transpuesta $\frac{V_2}{V_1} \Big|_{I_2=0}$. Se pretende, al usar los
ceros de transpuesta $\frac{V_2}{V_1} \Big|_{I_2=0}$, utilizarlos o eliminarlos focal o completo según
tipos de ceros finitos o infinitos.



32) Buses residuos por cada polo eliminados, o sea obtenemos nodos de componentes

$$Z_2 = Z_{22} - K_0' \cdot \frac{1}{s} \quad \text{donde } s = -1 \Rightarrow Z_2 = 0, K_0' = \text{Residuo de polo removido foradamente en origen}$$

$$Z_2|_{s=-1} = 0 = Z_{22}|_{s=-1} - K_0' \cdot \frac{1}{s} \Rightarrow 0 = \frac{s+2}{s(s+2.25)} - K_0' \cdot \frac{1}{s}$$

$$+K_0' \cdot \frac{1}{s} = \frac{s+2}{s(s+2.25)} \Rightarrow K_0' = \frac{s+2}{(s+2.25)} = \frac{-1+2}{-1+2.25} = \frac{1}{1.25} = \frac{4}{5} = K_0'$$

$$\text{Sabemos } K_0' \Rightarrow Z_2 = \frac{s+2}{s(s+2.25)} - \frac{4/5}{s} = \frac{(s+2) - (s+2.25) \cdot 4/5}{s(s+2.25)} = \frac{s+2 - 4/5 \cdot s - (4/5) \cdot 2.25}{s(s+2.25)}$$

$$Z_2 = \frac{\frac{1}{5}s + \frac{1}{5}}{s(s+2.25)} = \frac{(1/5) \cdot (s+1)}{s(s+2.25)} = Z_2 \quad \wedge \quad Y_2 = \frac{s(s+2.25)}{(1/5) \cdot (s+1)}$$

$$Y_4 = Y_2 - \frac{2K_1 \cdot s}{s+1} = \frac{s(s+2.25)}{(1/5)(s+1)} - \frac{2K_1 \cdot s}{s+1}, \quad 2K_1 = \lim_{s \rightarrow -1} Y_2 \cdot (s+1) = \lim_{s \rightarrow -1} \frac{s(s+2.25)}{(1/5)} = \frac{5}{4}$$

donde por \$ para ser admittancia y por lo que estamos

$$2K_1 = \lim_{s \rightarrow -1} \frac{s(s+2.25)}{(1/5)(s+1)} = \frac{(-1)(-1+2.25)}{(1/5)} = \frac{2.25}{4} = \frac{9}{16}$$

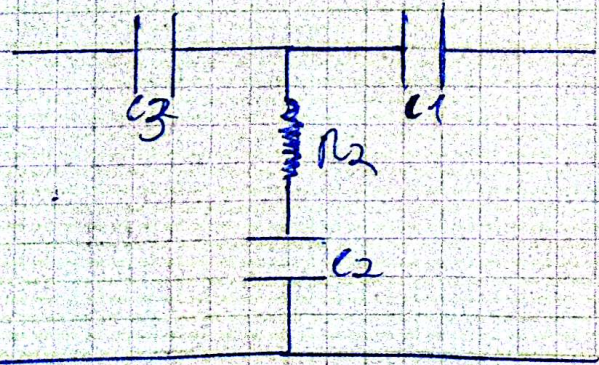
$$\Rightarrow Y_4 = \frac{s(s+2.25)}{(1/5)(s+1)} - \frac{(9/16)s}{s+1} = \frac{s(s+2.25) - (9/16)(s+1)}{(1/5)(s+1)}$$

$$Y_4 = \frac{s^2 + 2.25s - (9/16)s - 9/16}{(1/5)(s+1)} = \frac{s^2 + s}{(1/5)(s+1)} = \frac{s(s+1)}{(1/5)(s+1)} = 5s = Y_4 \quad \wedge \quad Z_4 = \frac{1}{5s}$$

$$Z_6 = Z_4 - \frac{K_2}{s}, \quad K_2 = \lim_{s \rightarrow 0} Z_4 \cdot s = \lim_{s \rightarrow 0} \frac{1}{5s} \cdot s = 1/5 = K_2$$

$$Z_6 = \frac{1}{5s} - \frac{1}{5s} = 0$$

42) Planteo circuito con nodos de componentes



$$C_1 = \frac{1}{K_0'} = 5/4, \quad C_3 = \frac{1}{K_2} = \frac{1}{1/5} = 5$$

$$C_2 = \frac{2K_1}{s} = \frac{25/4}{s} = 25/4$$

$$R_2 = \frac{1}{2K_1} = 4/25$$

2) Sintetice un circuito que cumpla con:

$$Z_{21} = \frac{s}{s^2+2} \quad y \quad Z_{22} = \frac{1+2s^2}{s(s^2+2)}$$

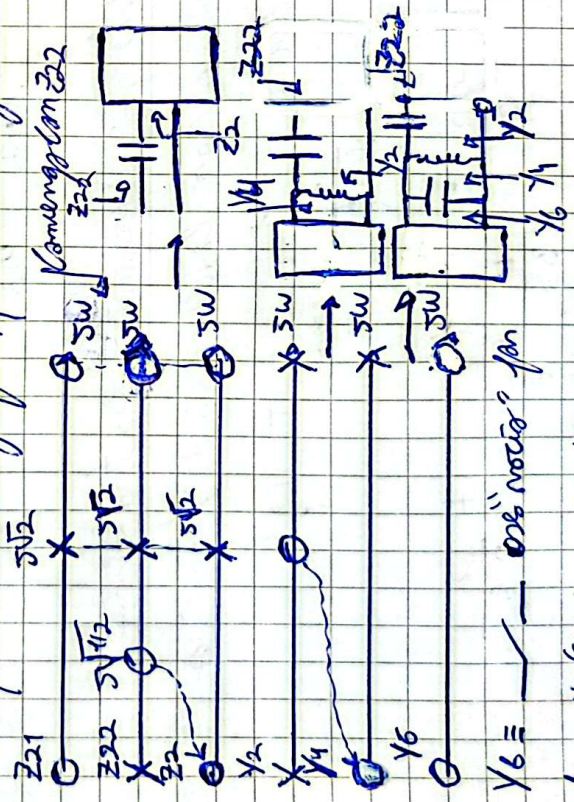
16) Observe estructuras de Z_{21} y Z_{22}

→ De $Z_{21} = \frac{s}{s^2+2}$, tiene "Zero" en $s=0$ y polo en $s = \pm\sqrt{2}$

→ De $Z_{22} = \frac{1+2s^2}{s(s^2+2)}$, tiene "Zero" en $s = \pm\sqrt{2}$ y polo en $s=0$ y $s = \pm\sqrt{2}$

Dado que hay polos y "Zeros" sobre eje SW y el origen $\Rightarrow$ síntesis por dispositivos

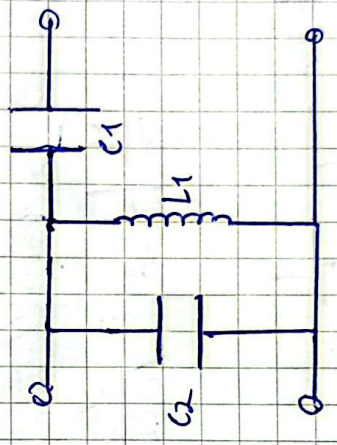
29) Aplique método gráfico, puede comenzar con Z_{21} o Z_{22} por interés, circuitos:



$Y_6 = \frac{1}{s}$ — "resistor" for

tenen "0" en origen y en SW con un SW

Circuitos Resultantes:



30) Puntos Nudosos por cada polo Nudo, así obtenga valores de componentes

$$Z_2 = Z_{22} - \frac{K_{01}}{s}, \quad K_{01} = \lim_{s \rightarrow 0} Z_{22}(s-0) = \lim_{s \rightarrow 0} \frac{1+2s^2}{s(s^2+2)} = \frac{1}{2} = K_{01}$$

$$Z_2 = \frac{1+2s^2}{s(s^2+2)} - \frac{1/2}{s} = \frac{1+2s^2 - (s^2+2)/2}{s(s^2+2)} = \frac{1+2s^2 - (s^2/2 + 1)}{s(s^2+2)} = \frac{(3/2)s^2}{s(s^2+2)} = \frac{3s}{2(s^2+2)}$$

$$Z_2 = \frac{(3/2)s}{s^2+2}, \quad Y_2 = \frac{s^2+2}{(3/2)s}$$

$$Y_4 = Y_2 - \frac{K_{02}}{s}, \quad K_{02} = \lim_{s \rightarrow 0} Y_2(s-0) = \lim_{s \rightarrow 0} \frac{(s^2+2)}{(3/2)s} = \frac{2}{3} = K_{02}$$

$$Y_4 = \frac{3s^2+2 - (4/3)}{(3/2)s}, \quad K_{02} = \frac{4}{3}$$

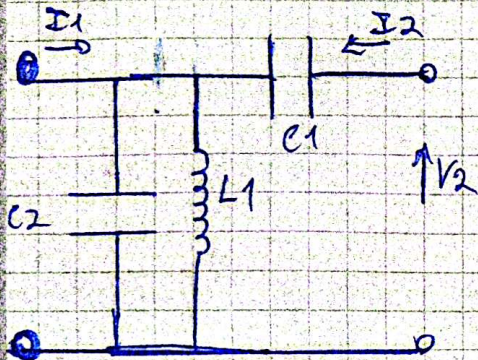
$$V_4 = \frac{\frac{2}{3}(s^2+2) - 4/3}{s} = \frac{\frac{2}{3}s^2 + \frac{4}{3} - \frac{4}{3}}{s} = \frac{2}{3}s$$

$$V_6 = V_4 - K_{00} \cdot s, \quad K_{00} = \lim_{s \rightarrow \infty} V_4 \cdot \frac{1}{s} = \lim_{s \rightarrow \infty} \frac{2}{3}s \cdot \frac{1}{s} = \frac{2}{3} = K_{00}$$

$$V_6 = \frac{2}{3}s - \frac{2}{3}s = 0$$

40) Plantas circuitos con volantes de componentes

circuitos con componentes



$$C_1 = \frac{1}{K_{01}} = \frac{1}{1/2} = 2$$

$$C_2 = K_{00} = \frac{2}{3}$$

$$L_1 = \frac{1}{K_{012}} = \frac{1}{4/3} = \frac{3}{4}$$

54) Realiza verificación del que obtenga el parámetro Z_{22} de los pds
Por esto se pueden plantear matrices o sino la definición de parámetros

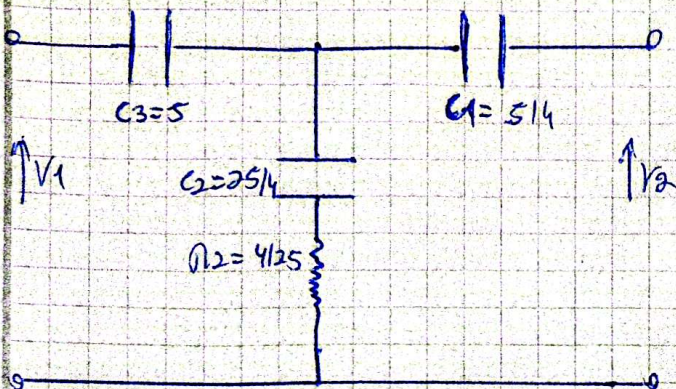
$$Z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0} = \frac{1}{Y_{L1/C2}} + Z_{C1} = \frac{1}{\frac{1}{\frac{1}{3}s} + K_{00}s} + \frac{K_{01}}{s}$$

$$Z_{22} = \frac{1}{\frac{1}{\frac{3}{4}s} + \frac{2}{3}s} + \frac{1/2}{s} = \frac{1}{\frac{4}{3s} + \frac{2}{3}s} + \frac{1}{2s}$$

$$Z_{22} = \frac{1}{\frac{4 + 2s^2}{3s}} + \frac{1}{2s} = \frac{3s}{4 + 2s^2} + \frac{1}{2s} = \frac{3s \cdot 2s + 2s^2 + 4}{(2s^2 + 4) \cdot 2s}$$

$$Z_{22} = \frac{8s^2 + 4}{(2s^2 + 4)2s} \cdot \frac{1/2}{1/2} \cdot \frac{1/2}{1/2} = \frac{2s^2 + 1}{(s^2 + 2)s} = Z_{22} \text{ Verifico } \checkmark$$

5º) Verificación



Ecuaciones primitivas T

$$V_1 = A \cdot V_2 + B (-I_2)$$

$$I_1 = C \cdot V_2 + D (-I_2)$$

$$\frac{1}{A} = \left. \frac{V_2}{V_1} \right|_{I_2=0} \quad / \quad \frac{1}{C} = \left. \frac{V_2}{I_1} \right|_{I_2=0}$$

Plantas producto matrices transpuestas

$$\begin{bmatrix} 1 & Z_{C3} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y_{C2} R_2 & 1 \end{bmatrix} \begin{bmatrix} 1 & Z_{C1} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+Z_{C3} Y_{C2} R_2 & 0+Z_{C3} \\ 0+Y_{C2} R_2 & 0+1 \end{bmatrix} \begin{bmatrix} 1 & Z_{C1} \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+Z_{C3} Y_{C2} R_2 + 0 & (1+Z_{C3} Y_{C2} R_2) Z_{C1} + Z_{C3} \\ Y_{C2} R_2 + 0 & Y_{C2} R_2 \cdot Z_{C1} + 1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

Resultado A

$$A = 1 + Z_{C3} Y_{C2} R_2 = 1 + \frac{1}{C_3} \cdot \frac{1}{\frac{1}{C_2} + R_2} = 1 + \frac{1}{\frac{C_3}{C_2} + C_3 R_2} \cdot \frac{1}{\frac{1}{C_2} + R_2} = 1 + \frac{1}{\frac{C_3}{C_2} + 5} \cdot \frac{1}{\frac{1}{25} + 5}$$

$$\frac{1}{C_2 R_2} = \frac{1}{\frac{25}{4} \cdot \frac{4}{25}} = 1, \quad \frac{1}{C_3 R_2} = \frac{1}{5 \cdot \frac{4}{25}} = \frac{5}{4}$$

$$\Rightarrow 1 + \frac{5/4}{5+1} = \frac{5+1+5/4}{5+1} = \frac{5+2.25}{5+1} = A$$

Resultado B

$$B = (1 + Z_{C3} Y_{C2} R_2) Z_{C1} + Z_{C3} = A Z_{C1} + Z_{C3} = \frac{5+2.25}{5+1} \cdot \frac{1}{\frac{C_1}{5/4}} + \frac{1}{\frac{C_3}{5}}$$

$$B = \frac{5+2.25}{(5+1)} \cdot \frac{(4/5)}{5} + \frac{1/5}{5} = \frac{(5+2.25)(4/5) + (5+1)/5}{(5+1) \cdot 5} = \frac{(4/5) \cdot 5 + (9/5) + (1/5) + (1/5)}{(5+1) \cdot 5} = \frac{5+2}{(5+1) \cdot 5} = B$$

Resultado 1/22

$$\frac{1}{V_{22}} = \frac{1}{B} = \frac{(5+2.25)}{(5+1)} \cdot \frac{(5+1) \cdot 5}{(5+2)} = \frac{(5+2.25) \cdot 5}{(5+2)} = 1/22, \text{ resultado}$$

Resultado 1/21

$$1/21 = -\frac{1}{D} \Rightarrow 1/21 = \frac{1}{\frac{5+2}{(5+1) \cdot 5}} = \frac{(5+1) \cdot 5}{(5+2)} = -1/21, \text{ resultado}$$